Speaker Practice Notes (Strict 5-Minute Delivery)

Bridging the Sim-to-Real Gap: Reinforcement Learning for Recommender Systems via *In Silico* LLM Personas

Brandon Tran (M.S. S&DS)
brandon.tran@yale.edu

October 20, 2026

Delivery Target: Strict 5-Minute Cap (≈625 spoken words total · 135–140 words/min · ∼4.5–4.6 min delivery).
Structure: Exact **1:1** match to each bullet on your slides. Bold text provides the visual anchor from the slide, integrated into a single, punchy spoken sentence with transitions.

Slide 1: Title Slide (∼20 seconds · 49 words)

Visual Scaffold: Title, Author, Yale S&DS Affiliation, Date.

- **On Netflix,** recommendations optimize for immediate clicks, ignoring that user satisfaction is an evolving sequential journey.
- **Our Thesis:** We use LLMs as *in silico* human personas—generative world models to safely train continuous RL without live user churn.
- **Transition:** Let's look at why current commercial recommenders struggle over longer horizons.

Slide 2: Grounding: Recommendation as Sequential Control (∼30 seconds · 73 words)

Visual Scaffold: What Recommenders Do Today (Myopic ranking) vs. What We Aim to Solve (Long-horizon sequential policies).

- **Today's platforms optimize for immediate clicks,** scoring items greedily while ignoring user state transitions $\mathcal{P}(s_{t+1} \mid s_t, a_t)$.
- **The Core Pitfall:** Repeated recommendations drive marginal utility to zero ($\frac{\partial U}{\partial n} \to 0$), causing genre fatigue and churn.
- **Sequential control** optimizes discounted return $\sum \gamma^t R_t$, trading a click today for sustainable retention next week.
- **The RL Dilemma:** Live exploration burns users, static logs suffer counterfactual bias, and toy simulators fail on humans.
- **Transition:** Existing workarounds try to address this, but suffer fundamental flaws.

Slide 3: Existing Paradigms & The Simulation Gap (∼35 seconds · 80 words)

Visual Scaffold: Simulators & Offline RL (RecSim, CQL) vs. Generative User Agents (Agent4Rec, The Missing Link).

- **Parametric Simulators (RecSim)** use rigid decay formulas; deep RL overfits to toy rules, failing on real humans.
- **Offline RL (CQL)** penalizes unseen actions, collapsing exploration across large catalogs into merely cloning logs.
- **LLM Agents** capture rich cultural priors, accurately mirroring real human rating distributions ($R^2 > 0.85$ on MovieLens).
- **The Missing Link:** LLMs were never used as interactive RL environments because text generation latency ($\sim 450\,\text{ms}$) prohibits training.
- **Transition:** We solve this by framing recommendation as a POMDP and eliminating text generation.

Slide 4: Proposal: In Silico Personas in a POMDP (∼40 seconds · 85 words)

Visual Scaffold: Recommendation POMDP Grounded on MovieLens (s_t, o_t) and Proposed Interactive Training Loop ($\boldsymbol{a}_t$, LLM step, R_t).

- **Empirical Grounding:** Grounded in real MovieLens histories, retaining authentic genres, timestamps, and ratings.
- **Hidden State (s_t):** Mood and fatigue are unobservable, requiring the policy to maintain an implicit belief state over history o_t.
- **Observed History (o_t):** The policy conditions strictly on observable profile embeddings and interaction sequences.
- **Continuous Action ($\boldsymbol{a}_t$):** Continuous vectors mapped via MIPS avoid discrete action branching, enabling efficient SAC optimization.
- **LLM Persona Step:** The persona evaluates the item, outputting a predicted rating and a session continuation flag ($c_t \in \{0, 1\}$).
- **Step Reward (R_t):** Balances immediate ratings with session retention, penalizing recommendations that exhaust users.
- **Transition:** Next, we validate persona fidelity before running policy training.

Slide 5: Method: Quantifying & Validating User Personas (∼35 seconds · 87 words)

Visual Scaffold: Proposed Behavioral Signatures (Genre preferences, satiation/fatigue, popularity) and Pre-RL Validation (Calibration, alignment, stability).

- **Genre Preferences** are grounded in empirical genre entropy $H(G) = -\sum p_g \log p_g$ from logs.

- **Satiation and fatigue** are captured organically via attention over history, without hand-coded formulas.
- **Popularity Tendency** models user preference for mainstream hits versus niche indie titles.
- **Rating Calibration** confirms simulated ratings match test distributions via calibration curves and MSE.
- **Distributional Alignment** measures preference overlap with real histories using Wasserstein distance.
- **Temporal Stability** confirms consistent rollouts across random seeds to prevent behavioral drift.
- **Transition:** Calibrated personas solve realism; now we tackle the computational inference bottleneck.

Slide 6: Fast Token-Logit Scoring & KV Caching ($\sim$35 seconds · 81 words)

*Visual Scaffold: Proposed Zero-Generation Token Scoring (Single pass formula) and Prefix Caching Across Parallel Archetypes (*100% *cache hit,* < 12 *ms latency).*

- **Zero-Generation Scoring:** Setting `max_tokens`=1 forces the rating as the next token, bypassing text generation entirely.
- **Exact Probabilities:** Softmax over logits gives a closed-form expected rating $\mathbb{E}[r_t]$ with zero parsing failures.
- **Archetype Clustering** allows parallel worker environments to share identical system prompts.
- **vLLM Prefix Caching** pins persona prefixes in GPU VRAM, eliminating redundant prompt computation.
- **Expected Throughput (< 12 ms):** Slashes latency by $> 35\times$, enabling 10^5 RL steps in 20 minutes on Bouchet HPC.
- **Transition:** This throughput enables a rigorous comparative benchmark.

Slide 7: Evaluation Protocol & Planned Experiments ($\sim$35 seconds · 86 words)

Visual Scaffold: Planned Three-Way Benchmark (RecSim vs. CQL vs. Generative SAC in $\mathcal{M}_{LLM}$*) and Planned Off-Policy Real Evaluation (Sim-to-real transfer, KuaiRec fallback).*

- **Parametric Baseline:** Continuous SAC in RecSim measures policy vulnerability to exploiting hand-crafted simulator rules.
- **Offline Baseline:** CQL benchmarks state-of-the-art offline RL trained directly on static logs.
- **Generative Setup:** Evaluates SAC trained against our high-throughput generative personas ($\mathcal{M}_{\text{LLM}}$).
- **Sim-to-Real Transfer** tests whether LLM training tightens the transfer error bound on held-out user trajectories.

- **KuaiRec Fallback:** A fully observed matrix serves as an uncorrupted counterfactual oracle if MovieLens exhibits high variance.
- **Transition:** Let's conclude with our core takeaways and roadmap.

Slide 8: Summary: Key Takeaways & Planned Roadmap (∼35 seconds · 85 words)

Visual Scaffold: Project Summary: What We Propose (Core problem, approach, key enabler) and Next Steps & Execution Plan (vLLM setup, baselines, SAC training).

- **The Core Problem:** Toy equations fail on real humans, creating a severe sim-to-real gap.
- **The Approach:** We recast recommendation as a POMDP where grounded LLM personas act as dynamic simulators.
- **Key Enabler:** Sub-12ms latency makes online continuous RL computationally practical on standard HPC hardware.
- **HPC Deployment:** First, deploy vLLM on Yale's Bouchet cluster and validate persona calibration curves.
- **Baselines:** Benchmark continuous SAC in RecSim against offline CQL.
- **Policy Evaluation:** Train SAC in $\mathcal{M}_{\text{LLM}}$ and measure sim-to-real transfer on real logs.